Notes on the Fast Multipole Method, Vortex Particle Method, and Neural Approximations

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I. Mathematical and Physical Foundations

A. Vector Calculus Review

Divergence: $\nabla \cdot \mathbf{F}$ measures the net flow out of a point. Gradient: $\nabla \phi$ gives the direction of steepest increase. Curl: $\nabla \times \mathbf{F}$ represents local rotation.

B. Integral Theorems (Gauss, Stokes, Green)

Gauss's theorem links divergence to flux through a surface. Stokes's theorem relates curl to circulation along a boundary. Green's identities provide integral relations for harmonic functions.

C. Inner and Outer Products

The dot product measures alignment; the cross product encodes area and orientation.

D. Potential Theory

Laplace's equation $\nabla^2 \phi = 0$ governs harmonic functions. Fundamental solutions and Green's functions are tools to construct solutions.

E. Numerical Analysis Essentials

Error, stability, and convergence define the quality of algorithms. Interpolation and quadrature enable discrete approximations of continuous functions. Computational complexity determines feasibility of large-scale simulations.

II. Fluid Dynamics Foundations

A. Vorticity and Velocity Formulations

The vorticity field $\omega = \nabla \times \mathbf{u}$ captures local rotation.

B. Biot–Savart Law

Velocity induced by a vortex filament can be written as

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \int \frac{\mathrm{d}\boldsymbol{\ell} \times (\mathbf{x} - \mathbf{x}')}{\|\mathbf{x} - \mathbf{x}'\|^3}.$$

C. Vortex Particle Method

Vorticity is discretized into particles with circulation strength. Particles advect with the flow and interact via smoothing kernels.

III. Fast Multipole Method (FMM)

A. Motivation: Reducing Complexity

Naive $O(N^2)$ interactions become $O(N)$ or $O(N \log N)$ with FMM.

B. Multipole Expansions

Laplace and Helmholtz kernels admit far-field expansions, reducing interaction cost.

C. Hierarchical Tree Structures

Quadtrees (2D) and octrees (3D) partition space into boxes, grouping particles adaptively.

D. Translation Operators

- M2M: Multipole to Multipole
- M2L: Multipole to Local
- L2L: Local to Local

E. Complexity Analysis and Error Control

FMM balances speed with controllable approximation error.

IV. FMM for the Vortex Particle Method

A. Coupling Biot–Savart with FMM

FMM accelerates evaluation of induced velocities from vortex particles.

B. Near-field vs. Far-field

Direct summation is used for nearby particles; multipole expansions handle distant ones.

C. Adaptive Trees

Nonuniform particle distributions require adaptive octrees for efficiency.

D. Accuracy and Kernel Regularization

Regularized kernels prevent singular behavior in particle interactions.

V. Neural Network Surrogates

A. Problem Framing

Inputs: relative position and circulation. Outputs: induced velocity vectors.

B. Network Architectures

Explored options include multilayer perceptrons (MLPs), graph neural networks (GNNs), and equivariant neural networks that respect rotational symmetry.

C. Training Data

Ground truth is generated from Biot–Savart computations. Supervised learning minimizes prediction error relative to true velocities.

D. Deployment Strategies

Neural surrogates replace kernel evaluations in FMM. Hybrid methods combine classical and neural approaches. Physical constraints are enforced to preserve conservation laws.

VI. Advanced Topics and Implementation

A. Error Control and Stability

Numerical stability is essential in long-time vortex simulations.

B. Parallelization and GPU Acceleration

Tree traversal and kernel evaluation map well to GPU architectures.

C. Real-Time Solver Design

Combining FMM and neural surrogates enables real-time vortex simulations.

D. Case Studies

Applications include vortex rings, wakes, and jet flows.

VII. References and Resources